

The Complete ML Engineer Playbook

Concept-wise YouTube playlist source pool

10 playlist-entry links per capability

How this works

These are not 150 playlists you must finish. They are a source pool for your AI ML Engineer Mentor. The mentor selects the smallest useful slice, explains it interactively, and converts it into experiments, builds, debugging and proof.

The rule

Never confuse resource consumption with capability. The mentor may use one playlist, several episodes from another, or ignore a source completely if it does not improve the current mission.

Important: Each link opens a YouTube playlist-focused search for that concept. Because YouTube playlists change, this is intentionally a live discovery layer rather than a frozen list of playlist IDs. The mentor should choose the strongest current result.

How the Mentor Uses This Library

The learner should not be forced to watch every source. The PDF supplies breadth; the mentor supplies sequencing, explanation and action.

MENTOR SOURCE-POOL INSTRUCTION

You are my ML Engineer Mentor. I am learning the capability: **[CAPABILITY]**.

Use the playlist source links supplied below as source material. Do not tell me to watch all of them. First give me a realistic engineering problem. Let me attempt it. Then identify the missing capability. Use the strongest relevant source material to explain only what I need, in short interactive episodes. Pause for predictions and questions. After understanding, make me implement something, break it, debug it, explain my choices, and prove the capability. If a source is redundant, skip it. If a source is outdated or weak, prefer a better current source. End with a concrete next mission.

Source strategy: the ten links under each capability are a maximum source pool. They are not a ten-course syllabus. The learner's actual path should usually be much smaller.

Python Foundations

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. Python programming beginner full course youtube.com / playlist search
02. Python programming fundamentals youtube.com / playlist search
03. Corey Schafer Python youtube.com / playlist search
04. freeCodeCamp Python youtube.com / playlist search
05. Bro Code Python youtube.com / playlist search
06. Real Python Python tutorial youtube.com / playlist search
07. Python functions data structures youtube.com / playlist search
08. Python modules files exceptions youtube.com / playlist search
09. Python problem solving youtube.com / playlist search
10. Python projects beginner youtube.com / playlist search

Mentor move → Start from the learner's current problem. Search these sources for the exact missing idea, explain it in an interactive episode, then move immediately to a practical experiment.

Object-Oriented Python

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. Python OOP classes objects
[youtube.com / playlist search](#)

02. Python object oriented programming
[youtube.com / playlist search](#)

03. Corey Schafer OOP Python
[youtube.com / playlist search](#)

04. Python inheritance polymorphism
[youtube.com / playlist search](#)

05. Python decorators dunder methods
[youtube.com / playlist search](#)

06. Python abstract classes interfaces
[youtube.com / playlist search](#)

07. Python design patterns
[youtube.com / playlist search](#)

08. Python clean code OOP
[youtube.com / playlist search](#)

09. Python advanced OOP
[youtube.com / playlist search](#)

10. Python OOP projects
[youtube.com / playlist search](#)

Mentor move → Start from the learner's current problem. Search these sources for the exact missing idea, explain it in an interactive episode, then move immediately to a practical experiment.

Git & GitHub

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. Git GitHub beginner youtube.com / playlist search
02. Git full course youtube.com / playlist search
03. Git branching merging youtube.com / playlist search
04. GitHub workflow youtube.com / playlist search
05. Git rebase cherry pick youtube.com / playlist search
06. Git advanced commands youtube.com / playlist search
07. Git pull request open source youtube.com / playlist search
08. Git conflict resolution youtube.com / playlist search
09. Git best practices youtube.com / playlist search
10. GitHub for developers youtube.com / playlist search

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NumPy Pandas SQL & Data

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. NumPy full course youtube.com / playlist search
02. Pandas full course youtube.com / playlist search
03. NumPy scientific computing youtube.com / playlist search
04. Pandas data cleaning youtube.com / playlist search
05. Pandas data analysis youtube.com / playlist search
06. SQL for data science youtube.com / playlist search
07. SQL advanced queries youtube.com / playlist search
08. Matplotlib Seaborn visualization youtube.com / playlist search
09. data cleaning preprocessing youtube.com / playlist search
10. exploratory data analysis EDA youtube.com / playlist search

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Math for ML

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. linear algebra machine learning youtube.com / playlist search
02. 3Blue1Brown linear algebra youtube.com / playlist search
03. calculus machine learning youtube.com / playlist search
04. derivatives gradients machine learning youtube.com / playlist search
05. probability for machine learning youtube.com / playlist search
06. statistics for machine learning youtube.com / playlist search
07. Bayes theorem machine learning youtube.com / playlist search
08. optimization machine learning youtube.com / playlist search
09. vectors matrices machine learning youtube.com / playlist search
10. math intuition machine learning youtube.com / playlist search

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Classical Machine Learning

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. machine learning full course youtube.com / playlist search
02. Stanford CS229 machine learning youtube.com / playlist search
03. MIT machine learning youtube.com / playlist search
04. Google machine learning crash course youtube.com / playlist search
05. scikit learn machine learning youtube.com / playlist search
06. regression classification machine learning youtube.com / playlist search
07. decision trees ensemble methods youtube.com / playlist search
08. unsupervised learning clustering PCA youtube.com / playlist search
09. model evaluation cross validation youtube.com / playlist search
10. feature engineering regularization youtube.com / playlist search

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ML From First Principles

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. linear regression from scratch youtube.com / playlist search
02. logistic regression from scratch youtube.com / playlist search
03. gradient descent from scratch youtube.com / playlist search
04. machine learning algorithms from scratch youtube.com / playlist search
05. neural network from scratch youtube.com / playlist search
06. backpropagation from scratch youtube.com / playlist search
07. decision tree from scratch youtube.com / playlist search
08. k means from scratch youtube.com / playlist search
09. PCA from scratch youtube.com / playlist search
10. machine learning mathematics from scratch youtube.com / playlist search

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Deep Learning

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. deep learning full course youtube.com / playlist search
02. MIT 6.S191 deep learning youtube.com / playlist search
03. Stanford CS231n youtube.com / playlist search
04. neural networks youtube.com / playlist search
05. deep learning optimization youtube.com / playlist search
06. CNN convolutional neural networks youtube.com / playlist search
07. RNN LSTM GRU youtube.com / playlist search
08. embeddings representation learning youtube.com / playlist search
09. deep learning regularization youtube.com / playlist search
10. deep learning from scratch youtube.com / playlist search

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PyTorch

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. PyTorch full course youtube.com / playlist search
02. PyTorch beginner youtube.com / playlist search
03. PyTorch neural networks youtube.com / playlist search
04. PyTorch training loops youtube.com / playlist search
05. PyTorch autograd tensors youtube.com / playlist search
06. PyTorch CNN youtube.com / playlist search
07. PyTorch NLP youtube.com / playlist search
08. PyTorch transformers youtube.com / playlist search
09. PyTorch projects youtube.com / playlist search
10. PyTorch debugging performance youtube.com / playlist search

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Transformers & LLMs

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. transformers from scratch youtube.com / playlist search
02. attention mechanism transformers youtube.com / playlist search
03. LLM fundamentals youtube.com / playlist search
04. large language models youtube.com / playlist search
05. Hugging Face transformers youtube.com / playlist search
06. tokenization embeddings LLM youtube.com / playlist search
07. LLM pretraining fine tuning youtube.com / playlist search
08. LoRA PEFT fine tuning youtube.com / playlist search
09. LLM inference quantization youtube.com / playlist search
10. LLM evaluation youtube.com / playlist search

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RAG & Modern AI Systems

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. RAG retrieval augmented generation youtube.com / playlist search
02. vector databases embeddings youtube.com / playlist search
03. LangChain RAG youtube.com / playlist search
04. LlamaIndex RAG youtube.com / playlist search
05. RAG evaluation youtube.com / playlist search
06. semantic search youtube.com / playlist search
07. hybrid search reranking youtube.com / playlist search
08. agentic RAG youtube.com / playlist search
09. LLM application engineering youtube.com / playlist search
10. modern AI systems youtube.com / playlist search

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Software Engineering for ML

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. software engineering Python youtube.com / playlist search
02. clean code Python youtube.com / playlist search
03. Python testing pytest youtube.com / playlist search
04. Python packaging youtube.com / playlist search
05. REST API FastAPI youtube.com / playlist search
06. software architecture Python youtube.com / playlist search
07. logging configuration Python youtube.com / playlist search
08. Docker Python youtube.com / playlist search
09. CI CD Python GitHub Actions youtube.com / playlist search
10. code review software engineering youtube.com / playlist search

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MLOps & Production

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. MLOps full course youtube.com / playlist search
02. MLflow MLOps youtube.com / playlist search
03. DVC data version control youtube.com / playlist search
04. Docker Kubernetes MLOps youtube.com / playlist search
05. model deployment machine learning youtube.com / playlist search
06. machine learning monitoring youtube.com / playlist search
07. ML CI CD youtube.com / playlist search
08. feature stores machine learning youtube.com / playlist search
09. model serving FastAPI youtube.com / playlist search
10. production machine learning systems youtube.com / playlist search

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Open Source ML Engineering

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. open source contribution GitHub youtube.com / playlist search
02. how to contribute to open source youtube.com / playlist search
03. GitHub issues pull requests youtube.com / playlist search
04. open source Python contribution youtube.com / playlist search
05. open source machine learning youtube.com / playlist search
06. reading large codebase youtube.com / playlist search
07. debugging open source projects youtube.com / playlist search
08. writing tests open source youtube.com / playlist search
09. maintainer workflow GitHub youtube.com / playlist search
10. first open source contribution youtube.com / playlist search

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ML Systems & Architecture

10 playlist-entry links • Use as mentor source material • Do not consume linearly

01. machine learning system design youtube.com / playlist search
02. ML system design interview youtube.com / playlist search
03. distributed machine learning youtube.com / playlist search
04. data pipelines machine learning youtube.com / playlist search
05. model serving architecture youtube.com / playlist search
06. feature store architecture youtube.com / playlist search
07. ML scalability latency cost youtube.com / playlist search
08. GPU distributed training youtube.com / playlist search
09. LLM systems architecture youtube.com / playlist search
10. production AI architecture youtube.com / playlist search

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What This Source Pool Is For

This library is the research layer behind the playbook—not the curriculum itself.

SOURCE POOL	10 playlist-entry links per capability
MENTOR	Selects, explains, questions, challenges and adapts
PRACTICE	Turns concepts into experiments and implementation
BUILD	Turns implementation into useful systems
BREAK	Introduces failure, debugging and trade-offs
PROOF	Requires evidence that the learner can do the work

Next layer: pair each capability with a direct AI Mentor chat link, then connect the learner to practical missions, real projects, open-source repositories, and real issues. The source pool supplies the knowledge; the mentor turns it into capability.